



10.4.4 Optimizing the tridiagonal QR algorithm

As the Givens' rotations are applied to the tridiagonal matrix, they are also applied to a matrix in which eigenvectors are accumulated. While one Implicit Francis QR Step requires $O(n)$ computation for chasing the bulge, this accumulation of the eigenvectors requires $O(n^2)$ computation with $O(n^2)$ data per step. This inherently means the cost of accessing data dominates on current architectures.

In a paper, we showed how accumulating the Givens' rotations for several Francis Steps before applying these to the matrix in which the eigenvectors are being computed allows one to attain high performance similar to that attained by a matrix-matrix multiplication.

- [47] Field G. Van Zee, Robert A. van de Geijn, Gregorio Quintana-Ortí, Restructuring the Tridiagonal and Bidiagonal QR Algorithms for Performance, ACM Transactions on Mathematical Software (TOMS), Vol. 40, No. 3, 2014.

For computing all eigenvalues and eigenvectors of a dense Hermitian matrix, this approach is competitive with the Method of Relatively Robust Representations (MRRR), which we mention in [Subsection 10.4.5](#)